

MATLAB Simulation Exercise

Statistical Signal Processing II

Kalman Filter

Group 09

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Theoretical Background

Many practical wireless communication channels are linear but time-varying, primarily due to multipath propagation and motion of scatterers in the environment. When sampled at the symbol rate, such channels can be accurately modeled as finite impulse response (FIR) filters with time-varying coefficients, commonly referred to as fading frequency-selective channels.

The discrete-time input–output relationship of a time-varying FIR channel with p taps is given by.

$$y[n] = \sum_{k=0}^{p-1} h_n[k] v[n - k] \quad (1)$$

where $v[n]$ is the transmitted signal and $h_n[k]$ the channel impulse response coefficient of the k –th path at time instant n . At the receiver end, a noise-corrupted version of the channel output is observed.

$$x[n] = \sum_{k=0}^{p-1} h_n[k] v[n - k] + w[n] \quad (2)$$

where $w[n]$ is additive white Gaussian noise (AWGN) with variance σ^2 .

In contrast to what we did in the first lab i.e. wiener filter where we were estimating a signal based on a desired signal, channel estimation aims to estimate the unknown and time-varying channel coefficients $h_n[k]$ with the assumption that the transmitted signal $v[n]$ is known.

By defining the channel coefficient vector as,

$$h[n] = [h_n[0] \ h_n[1] \ \dots \ h_n[p - 1]]^T$$

and the input vector $v[n] = [v[n] \ v[n - 1] \ \dots \ v[n - p + 1]]^T$

The observation $x[n]$ can be expressed as;

$$x[n] = v^T[n] h[n] + w[n] \quad (3)$$

To capture the slow temporal variation of the channel, the channel coefficients are modeled as a Gauss–Markov process, leading to the state equation below;

$$h[n] = Ah[n - 1] + u[n], \quad (4)$$

where A is the state transition matrix and $u[n]$ is zero-mean white Gaussian process noise with covariance matrix Q . Since each multipath component arises from different scatterers with independent motion, the corresponding channel taps evolve independently. Under the Gaussian assumption, this leads to statistically independent tap coefficients, which implies A , Q , and the initial covariance matrix can be chosen as diagonal matrices.

With the linear state equation (4) and linear observation equation (3) defined above, the channel estimation problem can be modelled using Kalman filtering framework. The Kalman filter provides a recursive minimum mean-square error (MMSE) estimator for the channel coefficient vector $h[n]$ by combining prior state predictions with noisy observations. At each time step n , the Kalman filter operates in two main phases i.e. prediction and correction [1].

In the prediction step, the filter propagates the previous state estimate and its error covariance forward in time using the state-space model. As a result, the predicted channel estimate is provided by;

$$\hat{h}[n|n-1] = A\hat{h}[n-1|n-1] \quad (5)$$

while the prediction error covariance,

$$M[n|n-1] = AM[n-1|n-1]A^T + Q \quad (6)$$

The Kalman gain which is used in the correction step is calculated as;

$$K[n] = \frac{M[n|n-1]v[n]}{\sigma^2 + v^T[n]M[n|n-1]v[n]} \quad (7)$$

In the correction phase, the prediction is updated using the new observation $x[n]$, weighted by the Kalman gain;

$$\hat{h}[n|n] = \hat{h}[n|n-1] + K[n](x[n] - v^T[n]\hat{h}[n|n-1]) \quad (8)$$

The updated error covariance is provided as:

$$M[n|n] = (I - K[n]v^T[n])M[n|n-1] \quad (9)$$

This recursive structure allows the estimator to track time-varying channel coefficients efficiently, adapting to changes while suppressing observation noise. When the transmitted signal is zero, the Kalman gain becomes zero, causing the filter to ignore purely noisy observations leading to robustness.

Problem Setup

In this laboratory, the Kalman filter is applied to estimate a two-tap time-varying channel. By varying the process noise covariance Q and the state transition matrix A , the impact of channel dynamics and modeling assumptions on estimation accuracy and tracking performance is investigated, as specified in the lab instructions. Below is a representation of the channel.

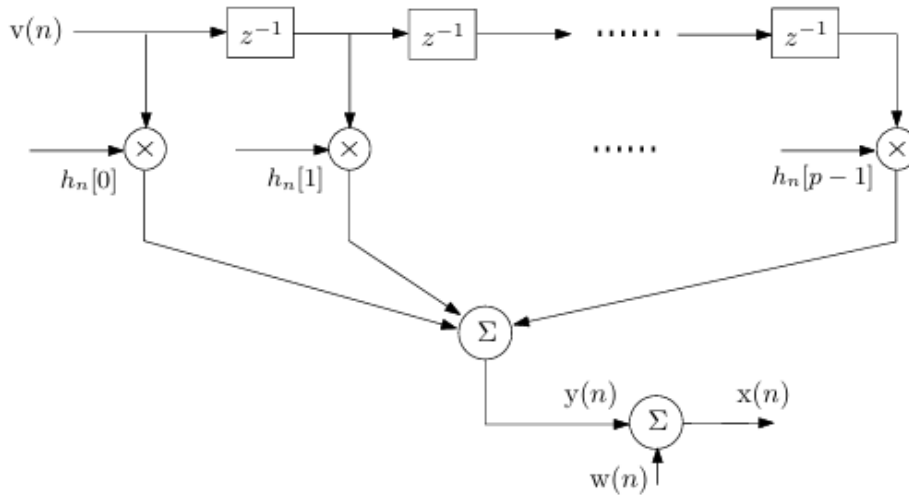


Fig. 1: FIR low-pass filter representation of the channel.

Task 1

In Task 1, we have implemented a Kalman filter for time-varying channel estimation. The channel is modeled as a linear, time-varying finite impulse response (FIR) filter whose tap weights evolve according to a Gauss–Markov process. Using a known input signal and noisy channel observations, the Kalman filter is applied to recursively estimate the channel tap coefficients. The objective of this task is to reproduce the reference results from Kay’s Example 13.3 and to illustrate the convergence, tracking behavior, and estimation accuracy of the Kalman filter under a slow-fading channel assumption.

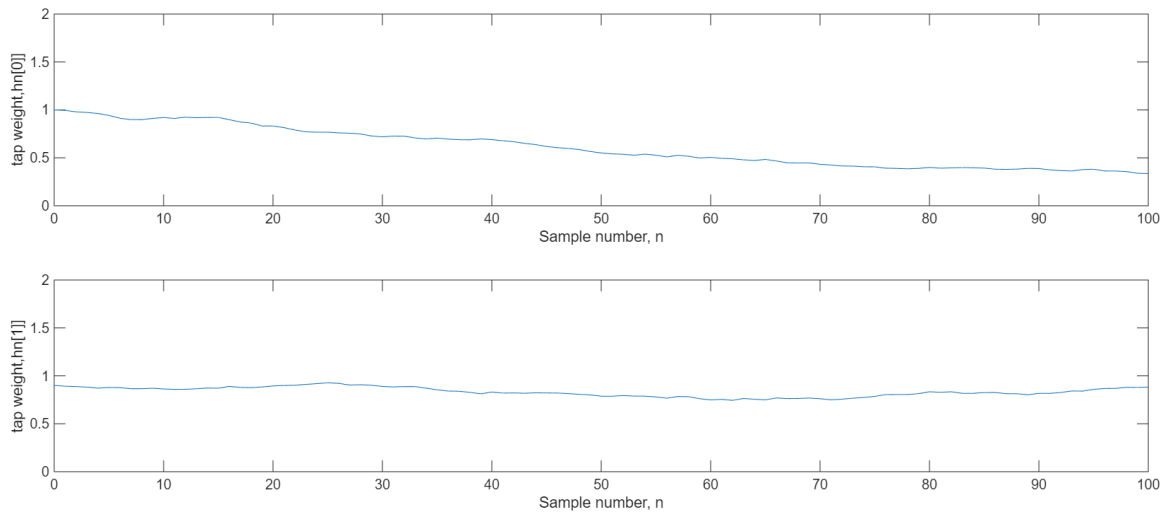


Figure 1.1 Realization of TDL coefficients

Figure 1.1 shows a realization of the time-varying channel tap weights $h_n[0]$ and $h_n[1]$. The first tap slowly decays over time, while the second tap remains approximately constant with small random fluctuations. This behavior is a direct consequence of the Gauss–Markov channel model, where the tap dynamics are governed by the state transition matrix A and process noise covariance Q . Since the diagonal entries of A are close to one, the channel varies slowly from sample to sample, representing a slow-fading channel. The random variations around the mean are caused by the process noise, which models small unpredictable changes in the propagation environment.

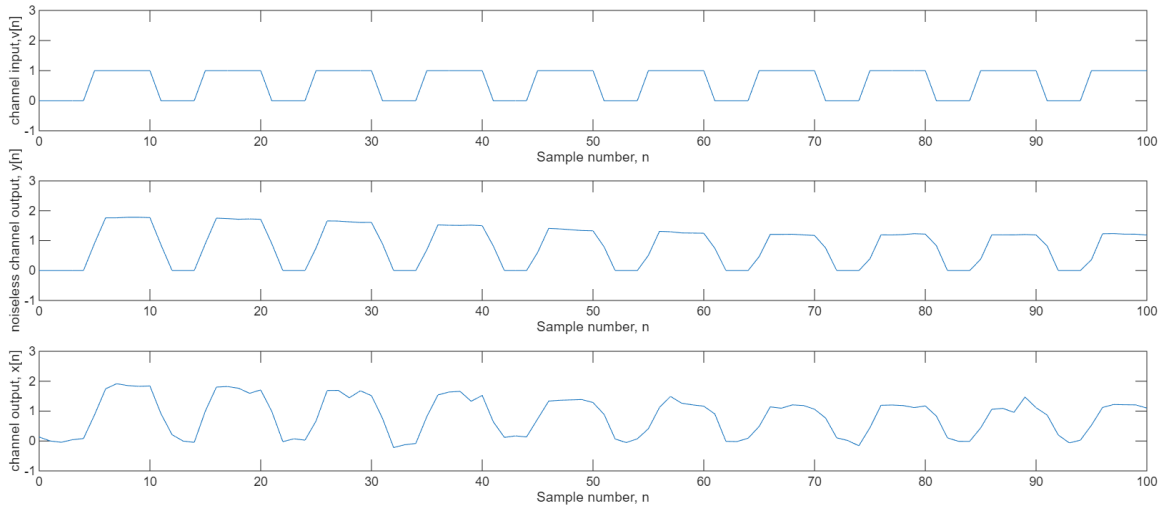


Figure 1.2 Realization of Input-Output waveforms of Channel

In figure 1.2 we have illustrated the input–output behavior of the time-varying channel. The top plot shows the transmitted input signal $v[n]$, which consists of a periodic sequence of pulses. The middle plot shows the noiseless channel output $y[n]$, where the signal shape is modified by the time-varying channel tap weights. As a result, the output amplitude changes slowly over time, reflecting the gradual variation of the channel. The bottom plot shows the received signal $x[n]$, which includes additive noise. Compared to the noiseless output, the noisy signal exhibits random fluctuations, but the underlying structure imposed by the channel is still clearly visible.

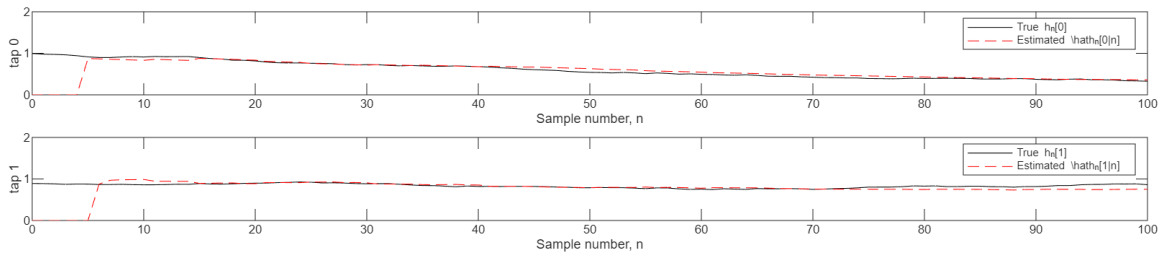


Figure 1.3 Realization of Kalman Filter Estimate

Figure 1.3 compares the true channel tap weights with their Kalman filter estimates over time. After a short initial transient, the estimated tap weights closely follow the true channel coefficients, indicating that the Kalman filter successfully locks onto the channel and tracks its slow variations. Small differences between the true and estimated curves are mainly due to observation noise and the continuous time variation of the channel. Overall, the results demonstrate that the Kalman filter provides accurate and stable channel estimates once the initial convergence phase has passed.

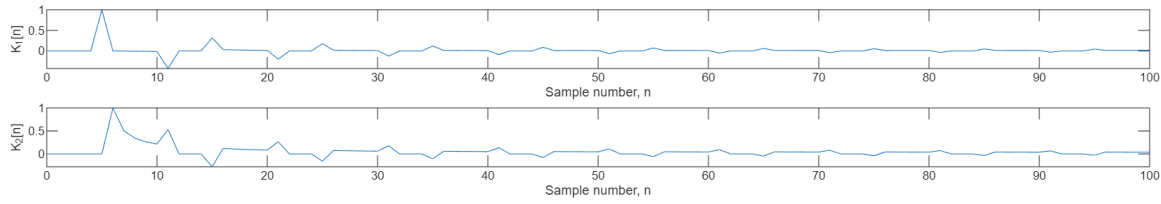


Figure 1.4 Realization of Kalman filter gains

In figure 1.4 we can see the Kalman filter gains for the two channel tap estimates. At the beginning, the gains are relatively large, indicating that the filter places strong emphasis on the incoming measurements due to the high initial uncertainty in the channel estimates. As the filter converges, the gains decrease and settle into a bounded, slowly varying pattern. Periods where the gains approach zero correspond to time instants when the input signal provides little or no information about the channel, causing the filter to rely mainly on its prediction model. This behavior reflects the adaptive nature of the Kalman filter and its ability to balance model predictions and noisy measurements.

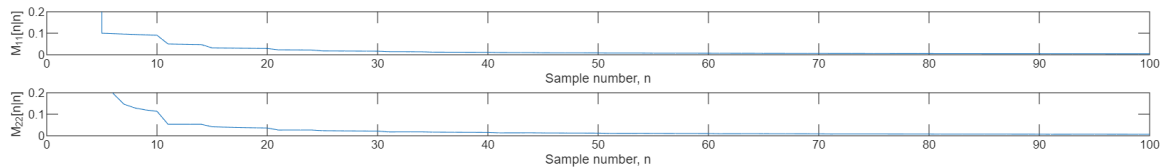


Figure 1.5 Realization of Kalman Filter minimum MSE

Figure 1.5 shows the evolution of the minimum mean-square error (MSE) for the two channel tap estimates. At the beginning, the MSE values are large, reflecting the high uncertainty in the initial channel estimates. As more observations become available, the Kalman filter rapidly reduces this uncertainty, leading to a sharp decrease in the MSE. After the initial convergence phase, the MSE settles at a small non-zero level, which represents the steady-state tracking error caused by measurement noise and the continuous time variation of the channel. This behavior confirms that the Kalman filter converges quickly and maintains stable tracking performance.

Task 2

In Task 2, the process noise covariance Q is increased while all other parameters remain unchanged. This change allows the channel tap weights to vary more rapidly over time, resulting in channel realizations that exhibit larger and more frequent fluctuations compared to Task 1. As a result, the noiseless and noisy channel outputs show stronger amplitude variations. In the Kalman filter results, the estimated tap weights still track the true channel coefficients, but with visibly increased tracking error and less smooth estimates. The Kalman gains remain larger on average, indicating that the filter relies more heavily on incoming measurements to follow the faster channel variations. Correspondingly, the minimum MSE curves converge to higher steady-state values, reflecting the increased uncertainty caused by the stronger process noise. Overall, increasing

Q makes the channel more dynamic and harder to track, which leads to reduced estimation accuracy and higher steady-state error.

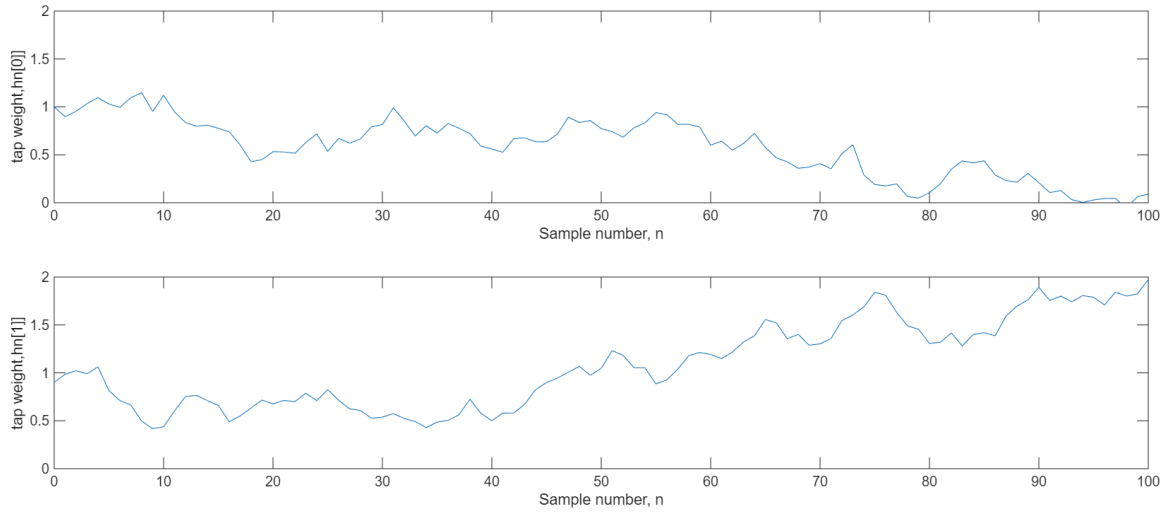


Figure 2.1 Realization of TDL coefficients

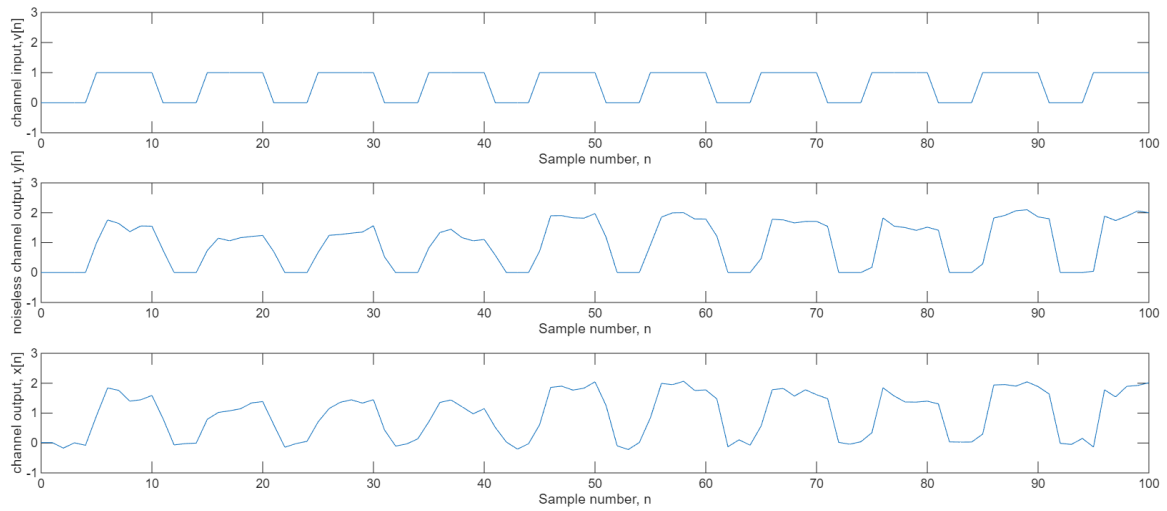


Figure 2.2 Realization of Input-Output waveforms of Channel

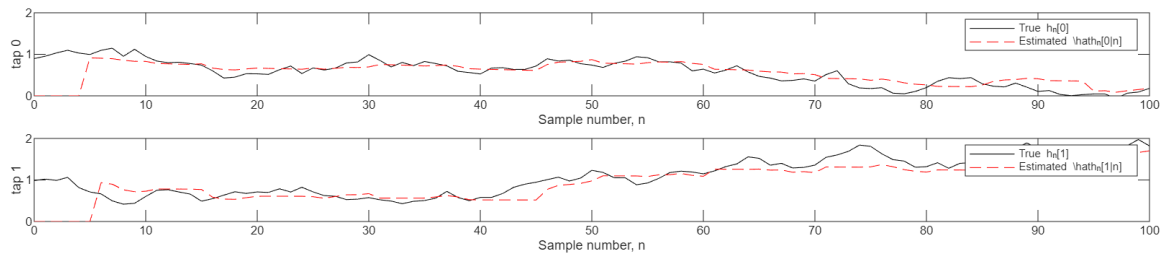


Figure 2.3 Realization of Kalman Filter Estimate

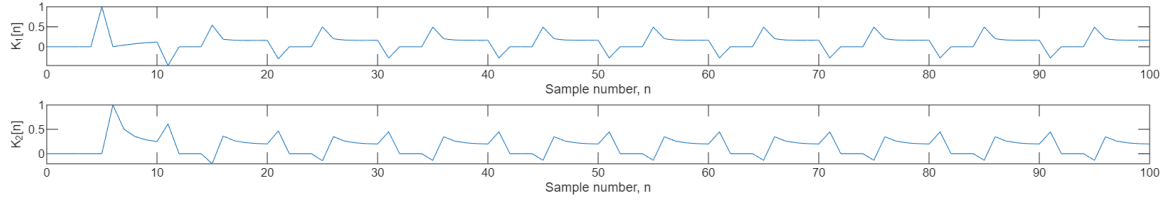


Figure 2.4 Realization of Kalman filter gains

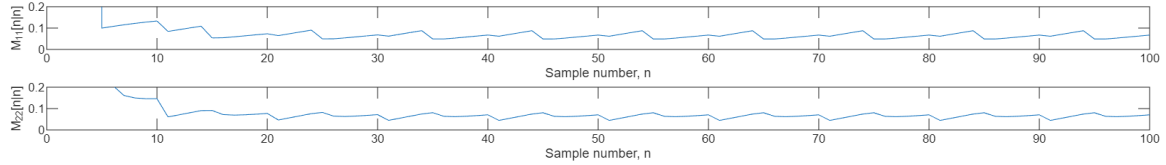


Figure 2.5 Realization of Kalman Filter minimum MSE

Task 3

In Task 3, the channel dynamics are modified by introducing off-diagonal elements in the state transition matrix A and by increasing the process noise covariance Q . The channel evolution is described by the Gauss–Markov state equation.

$$h[n] = Ah[n - 1] + u[n], \quad u[n] \sim N(0, Q)$$

$$A = \begin{bmatrix} 0.99 & 0.01 \\ 0.01 & 0.999 \end{bmatrix} \quad Q = \begin{bmatrix} 0.001 & 0 \\ 0 & 0.001 \end{bmatrix}$$

Compared to Tasks 1 and 2, the non-zero off-diagonal elements in A introduce coupling between the channel taps, meaning that the evolution of each tap depends not only on its own previous value but also on the previous value of the other tap. As a result, the channel coefficients become temporally and spatially correlated, leading to more complex channel trajectories. The increased values in Q further allow stronger random perturbations, enabling faster channel variations.

The measurement equation remains:

$$x[n] = v^T[n]h[n] + w[n]$$

$$w[n] \sim N(0, \sigma^2)$$

$$\text{where } v^T[n] = [v[n] \ v[n - 1]] \text{ and } \sigma^2 = 0.1$$

Due to the increased channel variability and tap coupling, the Kalman filter must rely more heavily on measurement updates. This is reflected in larger and more slowly decaying Kalman gains, given by eq. (7) where the prediction error covariance evolves as (6).

$$M[n | n - 1] = AM[n - 1 | n - 1]A^T + Q.$$

The increased Q directly raises $M[n | n - 1]$, resulting in higher steady-state values of the posterior error covariance. This behavior is observed in simulation results as higher minimum mean-square error (MSE) levels, less smooth tap estimates, and reduced tracking accuracy compared to the uncoupled case. Nevertheless, the Kalman filter remains stable and continues to track the channel successfully, demonstrating its robustness in the presence of correlated channel dynamics and increased process noise.

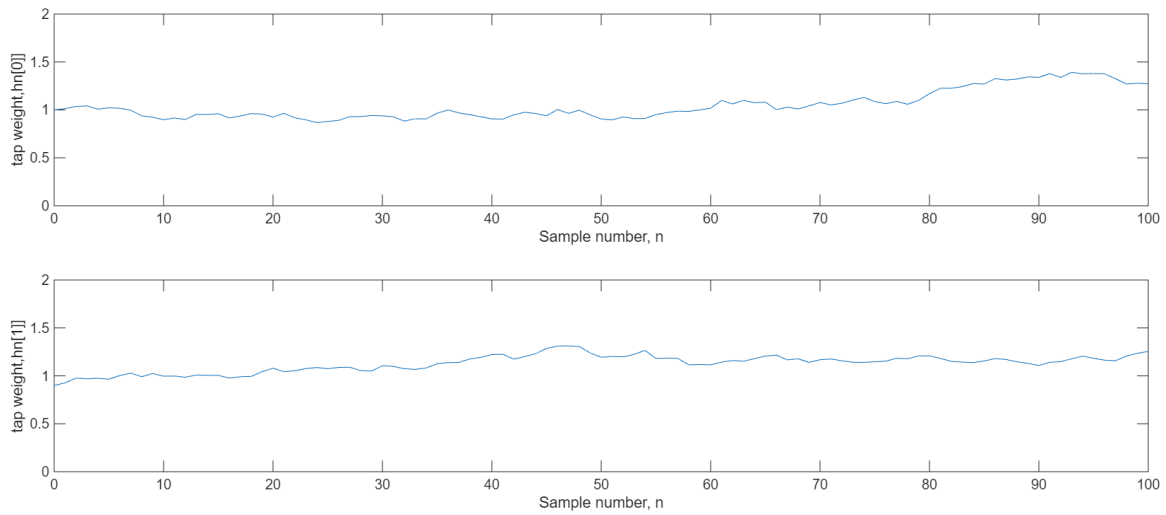


Figure 3.1 Realization of TDL coefficients

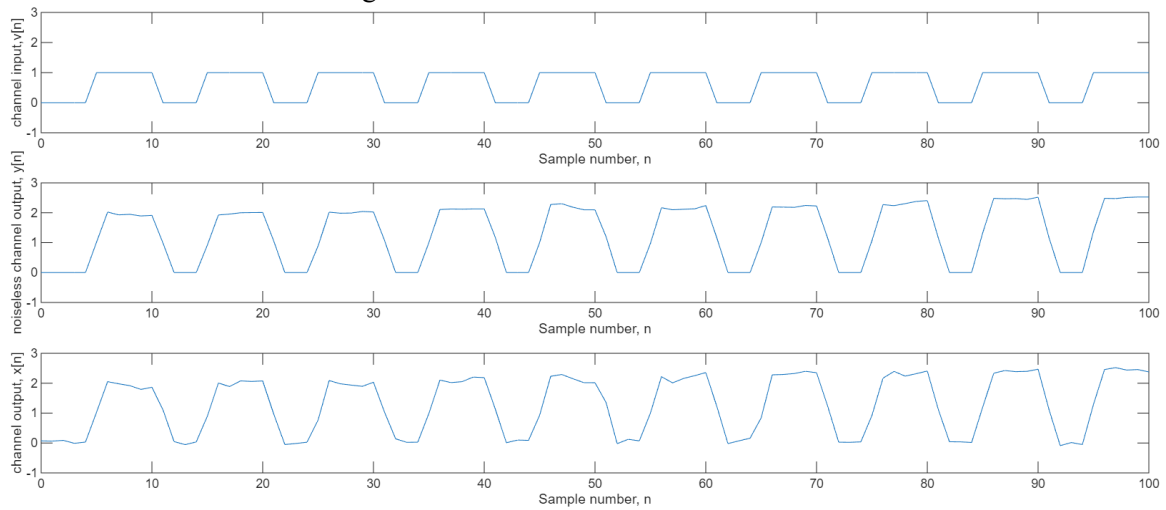


Figure 3.2 Realization of Input-Output waveforms of Channel

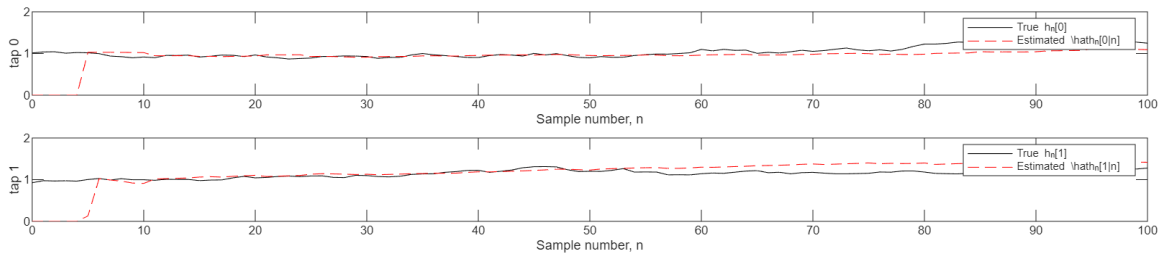


Figure 3.3 Realization of Kalman Filter Estimate

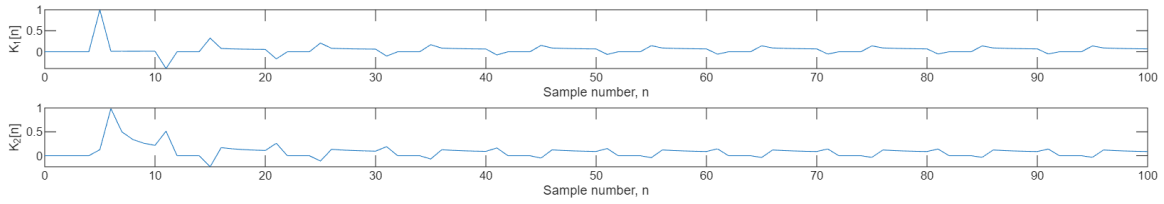


Figure 3.4 Realization of Kalman filter gains

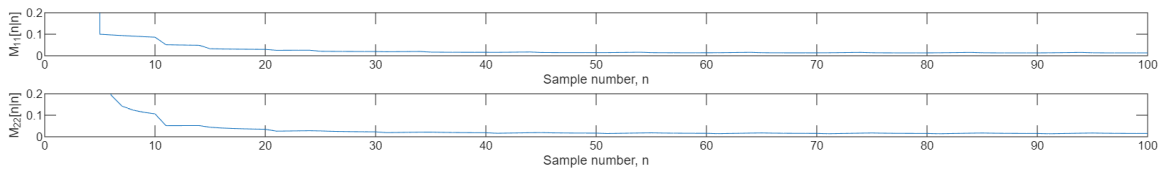


Figure 3.5 Realization of Kalman Filter minimum MSE

Conclusion

In this lab work, we studied the application of the Kalman filter to time-varying channel estimation and explored how different modeling assumptions affect estimation performance. By modeling the channel tap weights as a Gauss–Markov process, we learned how prior knowledge about channel dynamics can be incorporated into a recursive estimator. We have demonstrated through simulations how the Kalman filter combines model-based predictions with noisy measurements to achieve accurate channel tracking, even when only a single noisy observation is available at each time step.

Through the different tasks, we observed the impact of key parameters such as the state transition matrix and process noise covariance on the behavior of the filter. Increasing the process noise allowed faster channel variations but resulted in higher steady-state estimation error, while introducing coupling between channel taps led to correlated dynamics that were more challenging to track. These experiments highlighted the trade-off between model flexibility and estimation accuracy.

Overall, this exercise provided practical insight into how Kalman filters adapt to uncertainty and how tuning parameters influences convergence along with the reasoning careful modeling is crucial for reliable channel estimation in real-world systems.

REFERENCES

[1] S. M. Kay, Fundamentals of Statistical Signal Processing: Estimation Theory. Prentice Hall International, 1993.